Bereich: XML (2)

KML-Raster	**Musterlösung**
Package: `de.dhbwka.java.exercise.xml`	**Klasse:** `Grid`

```
package de.dhbwka.java.exercise.xml;

import java.io.File;
import java.io.FileOutputStream;
import java.io.OutputStreamWriter;
import javax.xml.parsers.DocumentBuilder;
import javax.xml.parsers.DocumentBuilderFactory;
import javax.xml.transform.OutputKeys;
import javax.xml.transform.Transformer;
import javax.xml.transform.TransformerFactory;
import javax.xml.transform.dom.DOMSource;
import javax.xml.transform.stream.StreamResult;
import org.w3c.dom.Document;
import org.w3c.dom.Element;

/**
 * Part of lectures on 'Programming in Java'. Baden-Wuerttemberg
 * Cooperative State University.
 *
 * (C) 2016 by W. Geiger, T. Schlachter, C. Schmitt, W. Suess
 *
 * @author DHBW lecturer
 * @version 1.0
 */
public class Grid {

	public static final int GRIDS = 5; // Rastergröße

	public static void main(String[] args) {
		Bounds kaBounds = Gazetteer.getBounds("Karlsruhe");
		try {
			DocumentBuilderFactory factory =
			    DocumentBuilderFactory.newInstance();
			DocumentBuilder parser = factory.newDocumentBuilder();
			Document doc = parser.newDocument();
			Element kml = doc.createElementNS(
			    "http://earth.google.com/kml/2.2", "kml");
			Element document = doc.createElement("Document");
			kml.appendChild(document);
			doc.appendChild(kml);
			int i = 0;
			double stepx = (kaBounds.east - kaBounds.west) / (GRIDS-1);
			double stepy = (kaBounds.north - kaBounds.south) / (GRIDS-1);
			for (double y = kaBounds.south; y <= kaBounds.north;
			      y += stepy)
			     for (double x = kaBounds.west; x <= kaBounds.east;
			          x += stepx) {
			       String coord = x + "," + y + ",0";
			       Element pm = doc.createElement("Placemark");
			       // Element "name" erzeugen und anhängen
```

```
                                Element name = doc.createElement("name");
                                name.setTextContent("Punkt " + i);
                                pm.appendChild(name);
                                // Element "description" erzeugen und anhängen
                                Element description =
                                    doc.createElement("description");
                                description.setTextContent("Koordinaten: " +
                                    coord);
                                pm.appendChild(description);
                                // Element "point" erzeugen und anhängen
                                Element point = doc.createElement("Point");
                                pm.appendChild(point);
                                // Element "point" erzeugen, an "point" anhängen
                                Element coordinates =
                                    doc.createElement("coordinates");
                                coordinates.setTextContent(coord);
                                point.appendChild(coordinates);
                                // neues "placemark" an kml anhängen
                                document.appendChild(pm);
                                i++;
                            }
                        TransformerFactory tFactory =
                            TransformerFactory.newInstance();
                        tFactory.setAttribute("indent-number", 2);
                        Transformer transformer = tFactory.newTransformer();
                        transformer.setOutputProperty(OutputKeys.INDENT, "yes");
                        DOMSource source = new DOMSource(doc);
                        StreamResult result = new StreamResult(
                                new OutputStreamWriter(new FileOutputStream(
                                    new File("karlsruhe.kml")), "UTF-8"));
                        transformer.transform(source, result);
                    } catch (Exception ex) {
                    }
            }

}
```

